

MSc Thesis Proposal

Parameter-Efficient Domain Adaptation of Foundation Models for Agricultural Vision

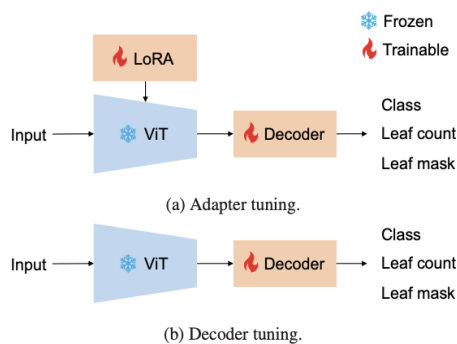
Contact: Vasileios Tzouras (vatzo@dtu.dk)

Lazaros Nalpantidis (lanalpa@dtu.dk)

Motivation

Large pretrained vision foundation models (e.g., DINOv2/v3, CLIP) are increasingly used as backbones for agricultural vision tasks, since they provide strong general-purpose features without requiring large labelled agricultural datasets. However, these models are pretrained on natural images and exhibit a domain gap when applied to field imagery, which differs in texture, occlusion, illumination, and fine grained visual similarity between classes. Adapting these backbones fully to every new farm, crop, or sensor is computationally expensive, and it remains unclear which parameter-efficient fine-tuning (PEFT) mechanism is best suited to closing this domain gap in agriculture, since prior work has mostly picked one PEFT method without comparing alternatives head-to-head on the same agricultural adaptation problem.

Keywords: Parameter-Efficient Fine-Tuning · Domain Adaptation · Precision Agriculture · Vision Foundation Models · Transfer Learning



Problem Statement

Given a pretrained foundation model and a target agricultural dataset, which PEFT mechanism best adapts the frozen backbone to that domain, and how does the answer change with the amount of target data available, the downstream task, or the domain shift involved? Existing agricultural adaptation studies typically fix a single PEFT method and a single dataset, leaving open which mechanism actually works best and why, and whether the best choice is

consistent across different types of agricultural domain shift.

Proposed Approach

The student will compare several PEFT methods for adapting one or two frozen foundation backbones (e.g., DINOv2/v3) to a set of publicly available agricultural datasets, keeping the task-specific head, trainable parameter budget, and training setup fixed across methods for a fair comparison. Public datasets that could be used include DeepWeeds (weed classification), CropAndWeed (crop/weed detection), PhenoBench (crop/weed segmentation), and GrowliFlower (cauliflower segmentation).

References to Get Started

- [1] [DINOv2: Learning Robust Visual Features Without Supervision](#)
- [2] [DINOv3](#)
- [3] [LoRA: Low-Rank Adaptation of Large Language Models](#)
- [4] [ExPLoRA: Parameter-Efficient Extended Pre-training to Adapt Vision Transformers Under Domain Shifts](#)
- [5] [Parameter Efficient Self-Supervised Geospatial Domain Adaptation](#)
- [6] [Enhancing Semantic Segmentation with Continual Self-Supervised Pre-training](#)
- [7] [Adapting Vision Foundation Models for Plant Phenotyping](#)